

Cem Ozcelik

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PROFESSIONAL SUMMARY

Data Scientist / ML Engineer with an Industrial Engineering background and nearly 5 years of progressive experience -- ~3.5 of which built directly inside an **automotive Tier-1 supplier** working on machine-sensor time-series and predictive maintenance. Deep stack across **Python, advanced SQL, statistical modelling, applied ML (XGBoost / LightGBM / scikit-learn / TensorFlow / PyTorch), anomaly detection, and distributed analytics on PySpark / Databricks**. Built a predictive maintenance and anomaly detection framework on CNC and stamping-machine telemetry that delivered a **15% weekly increase in critical asset availability** and a **15% OEE improvement** -- the same data shape and prognostics discipline that connected heavy-duty truck fleets require. End-to-end MLOps: containerized REST APIs in Docker / FastAPI / Django, full data quality and validation pipelines, and dashboards (Streamlit, Django) translating model decisions for non-technical reviewers. Industrial Engineering roots (MILP, AHP, cost-function design) make every ML problem a question of asymmetric cost-of-error first, accuracy second. Communicates clearly across cross-functional engineering, IT, and operations stakeholders in Turkish and English. Demo: github.com/ccemozclk/Credit-Risk | github.com/ccemozclk/Rossmann-Demand-Forecasting

CORE COMPETENCIES

Skills: Predictive Maintenance & Prognostics, Anomaly Detection, Time-Series Sensor Analytics, Statistical Modelling, Applied ML, Data Quality & Validation Pipelines, BI Dashboards & Reporting, Cross-Functional Engineering Collaboration (TR/EN)

Tools: Python (Advanced), SQL (Advanced), XGBoost, LightGBM, scikit-learn, TensorFlow, PyTorch, PySpark, Databricks, FastAPI, Django, Docker, Streamlit, AWS (EC2/S3), Git, Pandas / NumPy

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effe

Jan 2026 -- Present

Senior Database Analyst | Maltepe, Istanbul

- Took **full ownership of an end-to-end ML platform**, scaling an internal Django + Docker application into a hub serving versioned REST APIs for production analytics and ML models -- including data quality and validation checks at ingestion and inference time.
- Directed predictive supply chain analytics in Python and advanced SQL with statistical modelling and applied ML, sustaining an **8-hour weekly reduction in operational overtime**.
- Mined and analyzed internal databases to support product, optimize operational strategy, and improve business outcomes; translated outputs into **clear, actionable insights for technical and non-technical stakeholders**.
- Collaborated within cross-functional squads (engineering, operations, IT, external partners) to align analytics outputs with business needs and operational realities.
- **Tech:** Python, SQL, Django, REST APIs, Docker, PostgreSQL, ML Pipelines, Data Validation

Anadolu Pet Akvaryum & Effe

Sept 2024 -- Dec 2025

IT & Data Analytics Specialist | Maltepe, Istanbul

- Architected and deployed a full-stack Django web application from scratch (SOLID principles), establishing the foundation that later scaled into the production ML platform.
- Engineered predictive supply chain models in Python and PL/SQL that **reduced stockouts and improved inventory turnover**.
- Conducted Route-Level Profitability Analysis with advanced SQL, delivering a **3% improvement in overall logistics efficiency**.
- **Tech:** Python, Django, PL/SQL, Oracle, SOLID, full-stack deployment

Birinci Automotive (Tier-1 Automotive Supplier)

Jan 2022 -- Sept 2024

Mid-Level Data Analyst | Cayirova, Kocaeli

- Developed a **Predictive Maintenance & Anomaly Detection framework** on CNC and stamping-machine sensor time-series using unsupervised clustering on historical failure patterns -- the same prognostics discipline that connected vehicle and heavy-duty fleet telemetry require.
- Delivered a **15% weekly increase in critical machine availability** and a **15% improvement in Overall Equipment Effectiveness (OEE)**, validated by operations leadership on the factory floor.
- Engineered robust ETL pipelines in MSSQL and Python ingesting high-volume manufacturing and machine sensor data, with **validation rules and automated quality checks** replacing manual reporting workflows.
- Processed and analyzed complex time-series sensor data for cycle time optimization, building a strong foundation for sequential modelling and operational prognostics.
- Trained shift supervisors on alert interpretation, integrating ML outputs into preventive maintenance scheduling -- ML adoption with operational teams, not just model accuracy.
- Tech:** Python, MSSQL, ETL, unsupervised clustering, time-series, anomaly detection, predictive maintenance

Birinci Automotive (Tier-1 Automotive Supplier)

June 2021 -- Jan 2022

Junior Data Analyst | Cayirova, Kocaeli

- Conducted EDA and applied statistical methods to operational datasets, uncovering hidden patterns to optimize HR and production policies.
- Designed automated data workflows and ingestion pipelines, **improving reporting efficiency by 15%**.
- Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

End-to-End Credit Risk Prediction Anomaly Detection & MLOps

63% F1-Score with XGBoost on Lending Club, optimized for heavy class imbalance with temporal holdout validation (no leakage). Full MLOps pipeline: raw data ingestion -- feature engineering -- model training -- containerized production app in **Docker**. Treated class imbalance as a first-class design constraint -- compared SMOTE, undersampling, and class-weighted loss before settling on class-weighted XGBoost. Interactive EDA platform built with Django so non-technical reviewers can explore data and decisions. **Architecture and anomaly methodology transfer directly to vehicle prognostics and fleet anomaly detection** (FN cost >> FP cost in both domains).

Python, XGBoost, Django, Docker, Pandas, Scikit-learn | github.com/ccemozclk/Credit-Risk

Retail Demand Forecasting Time-Series Prognostics & Dashboard

~12.83% RMSPE on 6-week daily sales forecasting across 1,115 Rossmann stores using time-based validation. Object-oriented training pipeline (DataIngestion -- DataTransformation -- ModelTrainer) replacing static notebook code. **FastAPI REST inference + Streamlit dashboard with dynamic confidence intervals** translating ML metrics into business margin-of-error language -- directly analogous to a fleet-availability or remaining-useful-life dashboard for operational stakeholders. Resilient inference layer handling unseen categorical inputs in production.

Python, LightGBM, XGBoost, Optuna, FastAPI, Streamlit, Docker | github.com/ccemozclk/Rossmann-Demand-Forecasting

Scalable Recommendation Engine PySpark / Databricks at Scale

End-to-end personalized recommendation system on **Apache Spark / Databricks** for retail. **RFM analysis + K-Means** for behavioural segmentation; **ALS collaborative filtering** with implicit feedback and cold-start handling. Wrestled honestly with Spark-specific issues -- data skew, executor memory tuning, broadcast joins -- the same operational realities that show up when running connected-vehicle fleet analytics across millions of rows of telemetry on a distributed platform.

PySpark, Apache Spark, Databricks, Streamlit, K-Means, ALS | github.com/ccemozclk/Spark-Recommendation-Engine

EDUCATION

B.Sc. Industrial Engineering — Kirikkale University

Sept 2016 -- Feb 2021

Quantitative engineering degree (Mathematics, Statistics, Engineering core) with rigorous foundation in statistical analysis, operations research, and optimization. Designed MILP models with IBM ILOG CPLEX; applied AHP and TOPSIS to multi-criteria decision problems -- direct prerequisite for cost-function-aware ML design and prognostics trade-offs (False Negative vs False Positive in vehicle prognostics, predictive maintenance, and risk modelling).

CERTIFICATIONS

Associate Data Scientist Career Track (incl. PySpark, Recommendation Engines) — [DataCamp](#)

Jan -- May 2024

Data Science & Machine Learning Bootcamp (6 months, 10+ projects) — [Miuul](#)

Mar -- Sept 2022

SKILLS

Languages: Python (Advanced), SQL (Advanced), Bash, JavaScript (Basics)

Applied ML: XGBoost, LightGBM, scikit-learn, h2o.ai, Anomaly Detection, Predictive Maintenance, Time-Series, Statistical Modelling, A/B Testing, Feature Engineering

Deep Learning: CNN, RNN, LSTM, TensorFlow, Keras, PyTorch, Transfer Learning, Fine-Tuning, Grad-CAM (Explainable AI)

Big Data & Cloud: PySpark, Apache Spark, Databricks, Distributed Computing, AWS (EC2/S3)

MLOps & Deployment: Docker, FastAPI, Django, REST APIs, Git, Linux, End-to-End ML Deployment, CI-style retrain workflows

BI & Dashboards: Streamlit (production), Django EDA platforms, Dynamic Confidence Intervals; Power BI (ramp-ready)

Databases: PostgreSQL, MSSQL, Oracle, ETL Pipelines, Pandas, NumPy, Data Quality & Validation

Advanced Concepts: Cost-Function Optimization, MILP, AHP / TOPSIS, OOP, API & System Architecture, A/B Testing

Languages: Turkish (Native), English (Fluent)